

Atomic physics

Instructor:

Kaifeng Zhao (赵凯锋)

zhaokf@fudan.edu.cn,

18019208940 (Wechat)

Office: building 1, room 215

nuclear science and technology dept.



Study efficiently

- Flipped classroom
- Lecture notes and assignments centered
- Pay attention to new concepts and methods
- Use a pen and a notebook during the lecture
- Discussion
- Homework
 - submit on time
 - write your reasoning process, especially where you got stuck.

The Feynman Lecture on Physics

“If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generations of creatures, what statement would contain the most information in the fewest words?”

“I believe it is the atomic hypothesis that **all things are made of atoms** — little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another.”

Branches of Science heavily reply on atomic physics

- Astrophysics
- plasma physics
- atmospheric physics
- solid state physics
- chemical physics
- radiation physics
- Chemistry (analysis, reaction rates)
- biology (molecular structure, physiology)
- materials science
- energy research
- fusion studies

Modern atomic physics - Atomic, Molecular and Optical (AMO) Physics

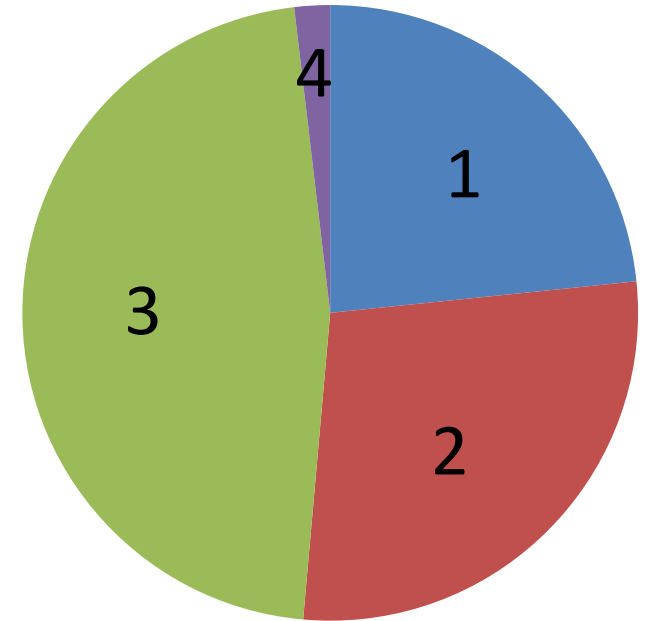
- Precision measurement
 - Testing the fundamental laws of physics
 - atomic clock, optical clock, nuclear clock
- Quantum sensing
 - Atomic magnetometer(磁力仪), atomic gyroscope (陀螺仪), atomic gravimeter (重力仪)
 - Magnetic Resonance Imaging (MRI) 核磁共振成像
 - Electromagnetic Induction Imaging (EMI) 电磁感应成像
- Cold atoms and ions, optical lattices
- Quantum information, computation and communication
- Ultra-intense and ultrafast lasers

We don't do atomic bombs, which should be called nuclear bombs.

We study the **electromagnetic** interactions between electrons, nucleus and external EM fields

Syllabus

1. The birth of atomic physics and quantum mechanics
2. elements of quantum mechanics
3. Atomic structure and interaction with EM fields
4. Advances in modern Advances



Birth of Atomic Physics and Quantum Mechanics

Key theoretical and experimental milestones, 1885–1928

● Atomic structure ● Atom–light ● Light ● Quantum theory

